

# CURRICULUM VITAE

## Alexandria K. Vail

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Human-Computer Interaction Institute  
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## Education

- 2016 – **Ph.D.**, Human-Computer Interaction  
Carnegie Mellon University
- 2015 – 2016 Ph.D. Student, Computer Science  
North Carolina State University  
Advisors: Dr. Kristy Boyer, Dr. James Lester
- 2011 – 2015 **B.S.**, Computer Science, **B.S.**, Mathematics, *magna cum laude*, 3.6/4.0  
North Carolina State University
- Minor concentrations: Cognitive Science, Physics  
Phi Kappa Phi, Sigma Pi Sigma  
University Honors, Computer Science Honors, Mathematics Honors

## Research Experience

- 2016 – **Graduate Research Assistant**, Human-Computer Interaction Institute  
Carnegie Mellon University
- 2015 – 2016 **Graduate Research Assistant**, Center for Educational Informatics  
North Carolina State University  
The LearnDialogue Group, The Intellimedia Group
- 2013 – 2015 **Undergraduate Researcher**, Center for Educational Informatics  
North Carolina State University  
The LearnDialogue Group

## Research Interests

ARTIFICIAL INTELLIGENCE, ADVANCED LEARNING TECHNOLOGIES, NATURAL LANGUAGE PROCESSING

- ARTIFICIAL INTELLIGENCE: user modeling, affective computing, AI & education
- ADVANCED LEARNING TECHNOLOGIES: intelligent tutoring systems, pedagogical agents, tutorial dialogue, game-based learning environments
- NATURAL LANGUAGE PROCESSING: natural language understanding, natural language dialogue systems

## Teaching Experience

- 2013 – 2014     **Teaching Assistant**, North Carolina State University  
Department of Computer Science
- Fall 2013        CSC 226: Discrete Mathematics for Computer Scientists  
Dr. Tiffany Barnes  
~250 students
- Spring 2014     CSC 230: C and Software Tools  
Dr. Sarah Heckman  
~120 students
- 2012 – 2013     **Mathematics & Physics Tutor**, North Carolina State University  
College Reading & Learning Association: Level II Advanced Tutor Certification

## Publications

### 2016

1. **A. Vail**, J. Grafsgaard, K. Boyer, E. Wiebe, and J. Lester. Gender Differences in Facial Expressions of Affect During Learning. *Proceedings of the Twenty-Fourth International Conference on User Modelling, Adaptation, and Personalization (UMAP 2016)*, pp. 65–73, Halifax, Canada, 2016. (long paper, acceptance rate: 23.9%)  
**Outstanding Paper**
2. **A. Vail**, J. Wiggins, J. Grafsgaard, K. Boyer, E. Wiebe, and J. Lester. The Affective Impact of Tutor Questions: Predicting Frustration and Engagement. *Proceedings of the Ninth International Conference on Educational Data Mining (EDM 2016)*, pp. 247–254, Raleigh, North Carolina, 2016. (long paper, acceptance rate: 27.5%)  
**Exemplary Paper**
3. W. Min, J. Wiggins, L. Pezzullo, **A. Vail**, K. Boyer, B. Mott, M. Frankosky, E. Wiebe, and J. Lester. Predicting Dialogue Acts for Intelligent Virtual Agents with Multimodal Student Interaction Data. *Proceedings of the Ninth International Conference on Educational Data Mining (EDM 2016)*, pp. 454–459, Raleigh, North Carolina, 2016. (short paper, acceptance rate: 52%)
4. **A. Vail**, J. Grafsgaard, K. Boyer, E. Wiebe, and J. Lester. Predicting Learning from Student Affective Response to Tutor Questions. *Proceedings of the Thirteenth International Conference on Intelligent Tutoring Systems (ITS 2016)*, pp. 154–164, Zagreb, Croatia, 2016. (long paper, acceptance rate: 15%)

### 2015

5. **A. Vail**, K. Boyer, E. Wiebe, and J. Lester. The Mars and Venus Effect: The Influence of User Gender on the Effectiveness of Adaptive Task Support. *Proceedings of the Twenty-Third International Conference on User Modeling, Adaptation, and Personalization (UMAP 2015)*, pp. 216–227, Dublin, Ireland, 2015. (long paper, acceptance rate: 28%)  
**Best Paper Award**

## 2014

6. **A. Vail**, J. Grafsgaard, J. Wiggins, J. Lester, and K. Boyer. Predicting Learning and Engagement in Tutorial Dialogue: A Personality-Based Model. *Proceedings of the Sixteenth ACM International Conference on Multimodal Interaction (ICMI 2014)*, pp. 255–262, Istanbul, Turkey, 2014. (long paper, acceptance rate: 39%)
7. J. Grafsgaard, J. Wiggins, **A. Vail**, K. Boyer, E. Wiebe, and J. Lester. The Additive Value of Multimodal Features for Predicting Engagement, Frustration, and Learning During Tutoring. *Proceedings of the Sixteenth ACM International Conference on Multimodal Interaction (ICMI 2014)*, pp. 42–49, Istanbul, Turkey, 2014. (long paper, acceptance rate: 39%)
8. **A. Vail** and K. Boyer. Adapting to Personality Over Time: Examining the Effectiveness of Dialogue Policy Progressions in Task-Oriented Interaction. *Proceedings of the Fifteenth Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL 2014)*, pp. 41–50, Philadelphia, Pennsylvania, 2014. (long paper, acceptance rate: 30%)
9. **A. Vail** and K. Boyer. Identifying Effective Moves in Tutoring: On the Refinement of Dialogue Act Annotation Schemes. *Proceedings of the Twelfth International Conference on Intelligent Tutoring Systems (ITS 2014)*, pp. 199–209, Honolulu, Hawaii, 2014. (long paper, acceptance rate: 17.5%)

## Honors and Awards

- Best Paper Award, Twenty-Third Conference on User Modeling, Adaptation, and Personalization, 2015.
- National Science Foundation Graduate Research Fellowship, 2015.
- North Carolina State University Department of Computer Science Senior Award for Scholarly Achievement, 2015.
- Generation Google Scholarship, 2014.
- North Carolina Society of Information Management Scholarship, 2013, 2014.
- Howard A. Petrea Endowed Scholarship in Mathematics, 2013.

## Relevant Coursework

### Graduate Coursework

Spoken Dialogue Systems  
Computational Applied Logic  
Numerical Analysis I  
Artificial Intelligence I, II

Software Engineering  
Combinatorics I  
Graph Theory

### Undergraduate Coursework

Computer Organization and Assembly  
Language for Computer Scientists  
C and Software Tools

Foundations of Advanced Mathematics  
Calculus I, II, III  
Applied Differential Equations I  
Introduction to Linear Algebra

Concepts and Facilities of Operating  
Systems for Computer Scientists  
Data Structures  
Automata, Grammars, and  
Computability  
Ethics in Computing  
Introduction to Artificial Intelligence

Introduction to Language & Linguistics  
Introduction to Cognitive Science  
Introduction to Psychology  
Cognitive Processes

Introduction to Modern Algebra for  
Mathematicians  
Introduction to Combinatorics  
Probability and Statistics for Engineers  
Mathematical Analysis I, II  
Mathematics of Scientific Computing

Technology & American Culture  
The Creative Process in Science  
Self, Schooling, & Social Order  
Introduction to College Tutoring

## Professional Membership

- Association for Computing Machinery (ACM).
- International Artificial Intelligence in Education Society (IAIED).
- Institute of Electrical and Electronics Engineers (IEEE).
- International Educational Data Mining Society (IEDMS).
- Students & Technology in Academia, Research, and Service (STARS) Student Leadership Corps.
- Mensa International.

## References

### Dr. Kristy Boyer

Associate Professor  
Department of Computer and  
Information Science and  
Engineering  
University of Florida  
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### Dr. James Lester

Distinguished Professor &  
Director of the Center for  
Educational Informatics  
Department of Computer Science  
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### Dr. Tiffany Barnes

Associate Professor  
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